ANCHORCOMPLETE: RECOVERING COMPLETE SCENE MESHES FROM PARTIAL POINT CLOUDS

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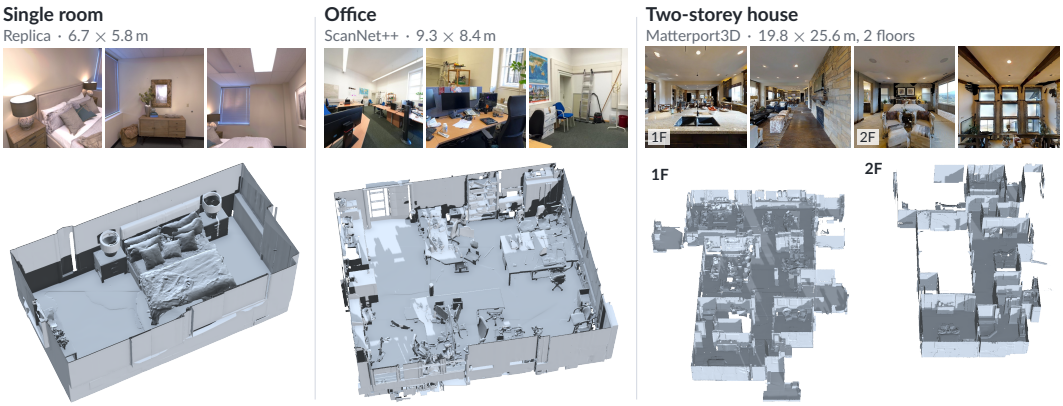


Figure 1: **ANCHORCOMPLETE from a single room to a building.** From the partial point cloud of a few posed views, ANCHORCOMPLETE recovers a complete, metric-scale scene mesh, shown for a room, an office and a two-storey house (depth of 10 views per floor; ceilings removed for display).

ABSTRACT

Most image-based 3D reconstructors prioritize reconstructing visible surfaces, producing partial point clouds with large holes and missing areas due to limited field of view and occlusions. To address this gap, we introduce ANCHORCOMPLETE, a framework that recovers complete scene meshes from partial point reconstructions. Our design addresses two key challenges: anchoring generation to the observed geometry and scaling completion to scenes of any extent. We introduce a *partial-point conditioning* module that guides both sparse-structure and sparse-latent generation using the input point cloud, together with *metric-scale coarse-to-fine generation*: an octree of fixed-size metric windows, each refining the layout of its coarser parent, keeps every mesh cell at $0.78\,\mathrm{cm}$ regardless of scene extent and recovers thin structures that a single normalized cube cannot resolve. We instantiate these designs on top of the sparse latent representation of TRELLIS. The resulting model produces complete scene meshes that remain faithful to the observed geometry, providing a more complete digital representation of the physical environment. Experiments on Replica and SCRREAM show that ANCHORCOMPLETE consistently outperforms strong baselines in complete scene reconstruction, particularly under sparse-view settings.

1 INTRODUCTION

We study the problem of recovering a complete 3D mesh of a scene given only a partial point observation, such as the output of a feed-forward 3D reconstructor or a 3D scanning system. A robust solution to this problem would benefit downstream applications such as any-view AR/VR rendering (Xie et al., 2026), large-scale physical simulation (Savva et al., 2019; Szot et al., 2021), and robotics (Liu et al., 2023a), where complete scene meshes are often required for rendering, collision detection, and physical interaction.

Addressing this problem requires tackling two coupled challenges: (i) preserving fidelity to the observed geometry and (ii) scaling completion to *scene*-level inputs. Recently, feed-forward 3D reconstructors like DUSt3R (Wang et al., 2024) and its variants (Leroy et al., 2024; Wang et al.,

2025b;a; Yang et al., 2025; Sun et al., 2024; Keetha et al., 2025), as well as online SLAM systems like DROID-SLAM (Teed & Deng, 2021) and VGGT-SLAM (Maggio et al., 2025), have made it possible to obtain dense point clouds of scenes in a single forward pass by scaling the model and training data to directly regress 3D points from image observations. However, completion is not part of their formulation, so unobserved regions remain large holes, limiting their applicability in many downstream applications.

Early completion methods (Han et al., 2017; Lyu et al., 2021; Yu et al., 2021; Zhou et al., 2021) focus on completing point clouds of single objects in normalized coordinate spaces and thus do not scale to scene-level inputs. While Amodal3R (Wu et al., 2025), SAM 3D (Team et al., 2025), and ShapeR (Siddiqui et al., 2026) extend foundational 3D generators (Zhang et al., 2023; Vahdat et al., 2022; Zhang et al., 2024; Hunyuan3D et al., 2025a; Xiang et al., 2024; 2025; Li et al., 2025a) to occluded geometry, they mainly address separate *object* completion in normalized coordinate spaces and do not directly condition on partial point observations, so they may not remain faithful to the geometry actually observed by the cameras. Recent *scene* completion methods like PaSCo (Cao et al., 2024), NOVA3R (Chen et al., 2026b), and Seen2Scene (Meng et al., 2026) aim to generate complete scene geometry. While promising, these models typically complete within the given field of view, without considering geometry outside the camera view. They also do not output a complete mesh, but rather a point cloud or a voxel grid, which may not be directly usable for downstream applications.

Seeking to close this gap, we introduce AnchorComplete, a generative framework that takes as input a partial point cloud, such as one recovered from sparse RGB observations, and outputs a complete scene mesh (Figure 1). We build on the sparse latent mesh representation of TRELLIS (Xiang et al., 2024; 2025), but adapting it to scene completion poses two key challenges: faithfully preserving observed geometry and scaling to scenes of any extent: inside one normalized cube the size of a mesh cell grows with the size of the scene, so the same grid that resolves a chair leg in a small room blurs it in an apartment.

We address these challenges with two main components. **Partial-point conditioning** anchors generation to the input through a hybrid scheme that combines global cross-attention to transformer-encoded anchor tokens with local spatial injection of observed voxels into the noisy latent. **Metric-scale coarse-to-fine generation** fixes the physical size of every cell and covers the scene with an octree of fixed-size metric windows, from $16\,\mathrm{m}$ windows that lay out the rooms to $4\,\mathrm{m}$ windows with $0.78\,\mathrm{cm}$ mesh cells; each finer window is generated with the coarser occupancy as a warm start in the regions no camera observed, so detail is added level by level while the room layout stays consistent across windows. Together, they enable faithful and scalable scene completion from sparse observations.

We evaluate our method on the Replica (Straub et al., 2019) and SCRREAM (Jung et al., 2024) datasets, where we reconstruct complete indoor scenes under standard and sparse-view settings. Our experiments demonstrate that AnchorComplete recovers more complete scene geometry than the baselines while remaining faithful to the visible input, and degrades more slowly as camera coverage drops. Our contributions are:

- We take a step towards recovering complete meshes of scenes from partial point observations, without requiring posed dense images, oriented normals, or per-scene optimization. For this new problem setting, we formulate strong baselines and evaluation protocols.
- We introduce AnchorComplete, a generative framework that contributes two architectural components: (1) **partial-point conditioning** and (2) **metric-scale coarse-to-fine generation** on an octree of fixed-size windows, which decouples mesh resolution from scene extent; together they enable generative mesh completion at scene scale while remaining anchored to the observed reconstruction.
- AnchorComplete achieves the best $\mathrm{CD}_{\mathrm{all}}$ and F-scores on Replica and SCRREAM, and degrades more slowly than the visible-surface baselines we evaluate as input views decrease.

2 Related Work

Visible-surface reconstruction. DUSt3R (Wang et al., 2024) and MASt3R (Leroy et al., 2024) regress dense pixel-aligned point maps from unposed image collections, marking a shift toward generalizable, pose-free 3D reconstruction. Building on this paradigm, several recent works (Wang

et al., 2025a;b; Lin et al., 2025; Yang et al., 2025; Sun et al., 2024; Keetha et al., 2025; Wang et al., 2025c; Chen et al., 2026a) extend it to multi-view settings, while online dense SLAM systems (Teed & Deng, 2021; Murai et al., 2025; Liu et al., 2025; Maggio et al., 2025) integrate such reconstruction into camera-trajectory estimation. Surface fitters such as Poisson and Screened Poisson reconstruction (Kazhdan et al., 2006; Kazhdan & Hoppe, 2013), and learned methods such as NKSR and NoKSR (Huang et al., 2023; Li et al., 2025b), convert oriented points into smooth implicit surfaces. All remain tied to camera-visible geometry: occluded regions go unmodeled, and surfaces fitted across holes follow smoothness rather than any prior over the unobserved scene. We instead complete those regions with a learned generative prior.

Generative 3D priors. Generative 3D methods synthesize complete geometry from images, text, layouts, or latent representations. Image-space methods such as Text2Room (Höllein et al., 2023), Invisible Stitch (Engstler et al., 2025), and GenRC (Li et al., 2024) iteratively inpaint and lift RGB-D views, while foundation 3D generators (Zhang et al., 2023; 2024; Xiang et al., 2024; Hunyuan3D et al., 2025a; Li et al., 2025a; Xiang et al., 2025) and amodal methods (Wu et al., 2025; Team et al., 2025) recover complete object geometry from image or text prompts. Scene-scale approaches (Shriram et al., 2025; Wu et al., 2024; Ju et al., 2024; Roessle et al., 2024; Bahmani et al., 2023; Yang et al., 2024a;b; Feng et al., 2023) generate rooms from learned scene representations or layouts. However, these priors are not directly constrained by the metric geometry of a specific partial reconstruction, motivating our point-conditioned formulation.

Completion from partial 3D geometry. Completion methods condition on partial 3D input to infer missing geometry. Object-centric approaches (Yuan et al., 2018; Yu et al., 2021; Zhou et al., 2022; Chu et al., 2023; Han et al., 2017; Lyu et al., 2021; Zhou et al., 2021) complete point clouds or shapes, while scene methods (Dai et al., 2018; 2020) predict occupancy or SDFs from partial voxel grids, typically at limited resolution. More recent generative models such as Hunyuan3D-Omni (Hunyuan3D et al., 2025b) and ShapeR (Siddiqui et al., 2026) condition foundation 3D generators on partial geometry but remain object-scale. At scene scale, PaSCo (Cao et al., 2024), NOVA3R (Chen et al., 2026b), Points-to-3D (Xia et al., 2026), Seen2Scene (Meng et al., 2026), and GenRecon (Schmid et al., 2026) explore occupancy, amodal reconstruction, latent generation, or chunk-based completion. Seen2Scene is the closest baseline to our setting and is evaluated in Section 4.2. Maintaining fidelity to observed geometry while plausibly completing missing regions remains a central challenge in scene completion. We condition a pretrained 3D generative prior on a raw scene-scale partial reconstruction and output a completed mesh in the original scene coordinate frame (Section 3).

3 METHOD

We develop ANCHORCOMPLETE, a generative model that recovers a complete scene mesh $\mathcal{M}$ from a partial point cloud $\mathcal{P}$ obtained from feed-forward 3D reconstruction (Wang et al., 2025a; Lin et al., 2025), SLAM (Teed & Deng, 2021; Maggio et al., 2025), or 3D scanning (Dai et al., 2017). The key challenge is to hallucinate plausible geometry in unobserved regions while preserving the observed structure. Our model, illustrated in Figure 2, builds on TRELLIS (Xiang et al., 2024) and adapts it to scene-level completion through partial-point conditioning and metric-scale coarse-to-fine generation. We first review TRELLIS in Section 3.1, then present partial-point conditioning in Section 3.2 and the coarse-to-fine octree that extends the generator from a single normalized cube to metric scale in Section 3.3, followed by training details in Section 3.4.

3.1 PRELIMINARIES: TRELLIS

TRELLIS (Xiang et al., 2024) is a foundational 3D asset generator that takes a text prompt or image as the condition $\mathbf{c}$ and outputs various 3D representations, including meshes $\mathcal{M}$, Gaussian splats $\mathcal{G}$, and radiance fields $\mathcal{R}$. For flexibility, TRELLIS represents the given 3D asset $\mathcal{O}$ with a set of *Structured Latents* (SLATs) $\mathcal{Z} = \{\boldsymbol{z}_i, \boldsymbol{p}_i\}_{i=1}^{L}$ on a predefined 3D spatial grid $\Omega \in \mathbb{N}^{H\times W\times D}$, where $L \ll HWD$ is the number of active voxels, $\boldsymbol{p}_i \in \mathbb{R}^3$ is the 3D coordinate of the i-th active voxel, and $\boldsymbol{z}_i \in \mathbb{R}^C$ is the corresponding latent feature encoding local geometry and appearance information.

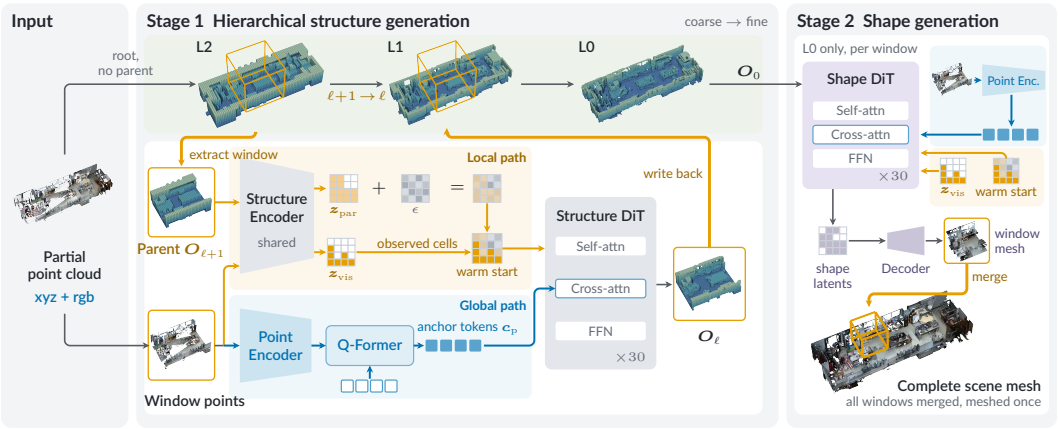


Figure 2: **Overview of AnchorComplete.** From a partial, colored point cloud, AnchorComplete generates a metric-scale octree coarse to fine, one window at a time (**Stage 1**), and then the shape latents at the finest level (**Stage 2**). Each window is conditioned on its points through anchor tokens (global path) and on its visible voxels and parent occupancy through the warm start (local path); the decoded windows are merged and meshed once.

The TRELLIS pipeline consists of two VAE modules followed by two corresponding rectified-flow generation stages (Albergo & Vanden-Eijnden, 2022; Lipman et al., 2022; Liu et al., 2023b). The first *structure* VAE $(\mathcal{E}_s, \mathcal{D}_s)$ compresses a dense binary voxel grid $\boldsymbol{O} \in \{0,1\}^{H\times W\times D}$ into a latent representation $\boldsymbol{z}_s \in \mathbb{R}^{\frac{H}{s}\times\frac{W}{s}\times\frac{D}{s}\times C}$, which is decoded back into a dense voxel grid to model coarse geometry. The second *SLAT* VAE $(\mathcal{E}_{sl}, \mathcal{D}_{sl})$ encodes the *Structured Latents* (SLATs) $\mathcal{Z}$, augmented with voxelized multi-view image features of the 3D asset, into a latent representation $\boldsymbol{z}_{sl} \in \mathbb{R}^{L\times C}$, which is then decoded into various 3D representations. During generation, two rectified-flow stages are applied to the structure and SLAT latent spaces, respectively. In the first sparse structure generation stage, a flow transformer $\mathcal{T}_s$ generates the structure latent $\boldsymbol{z}_s$ from Gaussian noise conditioned on the text or image prompts $\mathbf{c}$. The sampled $\boldsymbol{z}_s$ is then decoded into the active voxel support $\{\boldsymbol{p}_i\}_{i=1}^L$, which determines the coarse spatial layout of the asset. In the second SLAT generation stage, a sparse flow transformer $\mathcal{T}_{sl}$ generates the local SLAT features $\{\boldsymbol{z}_i\}_{i=1}^L$ on this support $\{\boldsymbol{p}_i\}_{i=1}^L$, again conditioned on $\mathbf{c}$. Finally, the generated SLAT $\mathcal{Z}$ is decoded into the desired 3D output. In our work, we adopt the TRELLIS SLAT representation and extend it with partial-point conditioning and scalable scene-level generation.

3.2 Partial Point Conditioning

Unlike image-conditioned 3D generators (Zhang et al., 2024; Hunyuan3D et al., 2025a; Xiang et al., 2024; 2025), we condition scene completion directly on partial point observations. This is advantageous for several reasons: (1) *Completeness*: the observed points delineate the visible scene boundary, providing direct geometric evidence of where completion is needed; (2) *Accuracy*: the condition is directly grounded in 3D geometry, maintaining better fidelity to the visible geometry; (3) *Efficiency*: the observed points $\mathcal{P} = \{\boldsymbol{p}_i\}_{i=1}^N$ can be uniformly subsampled to a fixed size N, avoiding conditioning costs that grow with the number of input images; and (4) *Composability*: point-based inputs can be chunked for scalable local processing, and the outputs can be directly merged in the same scene coordinate frame. In particular, we take a hybrid conditioning approach via (1) global cross-attention to *anchor tokens* encoded from the point cloud and (2) *local spatial injection* of visible-voxel evidence into the structure latent; both paths are shown in Figure 2. The two pathways are ablated in Section 4.

Global Anchor Tokens. We design a simple architecture to extract anchor tokens from the input point observations $\mathcal{P} = \{\boldsymbol{p}_i\}_{i=1}^N$. In particular, we implement the point-cloud encoder $\mathcal{E}_p$ using a transformer architecture inspired by the point encoder of NOVA3R (Chen et al., 2026b). To this end, we first map the 3D coordinates of all N input points to high-dimensional features $\{\boldsymbol{f}_i\}_{i=1}^N$. The encoder then selects $\mathcal{F}^q = \{\boldsymbol{f}_i\}_{i=1}^m$ point queries via farthest point sampling and uses the full set of point features $\mathcal{F}^{(k,v)} = \{\boldsymbol{f}_i\}_{i=1}^N$ as the keys and values for the encoder's cross-attention layers,

where $m \ll N$ is a fixed number of point tokens. The anchor tokens are further refined by several interleaved self-attention and cross-attention blocks. The resulting anchor tokens $\boldsymbol{c}_{\text{p}} \in \mathbb{R}^{m \times C_{\text{xa}}}$, where C_{xa} is the cross-attention feature dimension, are then fed to the TRELLIS flow model as global conditioning tokens through cross-attention, replacing the original image/text conditioning tokens in both stages. The metric-scale model of Section 3.3 instead uses a pretrained Utonia point backbone (Zhang et al., 2026) and a Q-Former that compresses its features into $m = 768$ anchor tokens, reading point colours, an observed-point flag and the expected depth noise.

Local Spatial Injection. To further encourage spatial consistency between the generated geometry and the input point cloud, we introduce a local spatial conditioning path that injects visible-point evidence directly into the first-stage structure latent space. Inspired by condition injection strategies in image editing (Avrahami et al., 2023), we first voxelize the point observations $\mathcal{P}$ into a binary 3D grid $\boldsymbol{O}_{\text{vis}} \in \{0,1\}^{H \times W \times D}$, setting a voxel to 1 if it contains at least one observed point and to 0 otherwise. It is then encoded with the pretrained structure encoder $\mathcal{E}_{\text{s}}$ to obtain partial-geometry features $\boldsymbol{z}_{\text{vis}} \in \mathbb{R}^{\frac{H}{s} \times \frac{W}{s} \times \frac{D}{s} \times C}$. In parallel, we downsample $\boldsymbol{O}_{\text{vis}}$ to obtain a 3D binary mask $\boldsymbol{O}_{\text{vis},m} \in \{0,1\}^{\frac{H}{s} \times \frac{W}{s} \times \frac{D}{s}}$, indicating which latent cells correspond to visible regions. The visible evidence enters Stage 1 in two ways. First, it initializes the flow: rather than starting from Gaussian noise everywhere, we replace the initial sample at visible cells by $\boldsymbol{z}_{\text{vis}}$,

$$\boldsymbol{z}_{\text{s}}^{(1)} \;=\; \boldsymbol{O}_{\text{vis},m} \odot \boldsymbol{z}_{\text{vis}} \;+\; (\mathbf{1} - \boldsymbol{O}_{\text{vis},m}) \odot \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}), \tag{1}$$

where $\boldsymbol{z}_{s}^{(1)}$ is the latent the sampler starts from. Second, the mask is available at every denoising step: we concatenate $\boldsymbol{O}_{\text{vis},m}$ to the noisy latent and feed the result back to the denoiser, so it can separate observed from unobserved cells throughout sampling. Both act only in Stage 1; we do not apply them to Stage 2, since Stage 1 already predicts the complete active voxel support on which SLAT features are generated. The TRELLIS.2-based models of Section 3.3 apply both to Stage 2 too, with the shape latents of the visible points and a visibility flag at every step and as the initial sample on observed voxels.

3.3 Metric-Scale Coarse-to-Fine Generation

A single normalized cube ties resolution to scene extent: scaled to enclose the input, its mesh grid has cells several centimetres wide on a large apartment, too coarse for chair legs, shelf boards and door frames, and finer than needed on a small room. We instead fix the cell size in metres and cover the scene with as many fixed-size windows as it needs, organized as an octree whose coarse levels decide the layout and whose fine levels add detail.

Metric octree. All levels share one gravity-aligned lattice in metric world coordinates. Level $\ell \in \{0,1,2\}$ tiles space with cubic windows of side $B_\ell = 4 \cdot 2^\ell$ m, each holding a 64^3 structure grid and a 512^3 mesh grid of cell $c_\ell = 2^\ell/128$ m (0.78 cm at $\ell = 0$, 3.1 cm at $\ell = 2$). Both stages of Section 3.2 run unchanged inside each window on its crop of the input, so a window looks the same to the model at every level and in every scene. We use the sparse-structure and O-Voxel shape latents of TRELLIS.2 (Xiang et al., 2025).

Coarse-to-fine generation. The root, level 2, comes from one pass of the conditioned generator over a single cube enclosing the scene, voxelized onto the level-2 structure grid; it supplies the whole-scene layout. Each finer level ℓ places windows only over the occupied region of level $\ell{+}1$ and is conditioned on the input points, through the anchor tokens and visible-voxel injection of Section 3.2, and on the parent occupancy $\boldsymbol{O}_{\ell+1}$, encoded as $\boldsymbol{z}_{\text{par}} = \mathcal{E}_{\text{s}}(\boldsymbol{O}_{\ell+1})$, which seeds the unobserved cells of the initial sample,

$$\boldsymbol{z}_{s}^{(1)} \;=\; \boldsymbol{O}_{\text{vis},m} \odot \boldsymbol{z}_{\text{vis}} \;+\; (\mathbf{1} - \boldsymbol{O}_{\text{vis},m}) \odot \big(\alpha\, \boldsymbol{z}_{\text{par}} + \sqrt{1-\alpha^2}\, \boldsymbol{\epsilon}\big), \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}), \tag{2}$$

which reduces to Equation (1) for $\alpha = 0$; we use $\alpha = 0.8$, so the finer level refines the parent's layout instead of regenerating it. Generated occupancy is kept only inside the parent dilated by one parent cell or next to an observed point, so every child lies inside its parent; windows overlap by a quarter of their side and keep only their centre, so neighbours meet without seams. Stage 2 then generates shape latents at the finest level, and the decoded windows are merged on the level-0 lattice and remeshed once; decoding a coarser level instead yields the same scene at 3.1 or 1.6 cm cells. Detail is thus independent of scene extent and memory grows with the occupied volume, while each window solves a local refinement that a 64^3 generator can handle and the coarse levels keep the layout consistent across windows.

3.4 TRAINING

We fine-tune two flow models in TRELLIS with our new conditioning: one predicts the active voxels $\{\boldsymbol{p}_i\}_{i=1}^{L}$, and the other predicts the SLAT features $\{\boldsymbol{z}_i\}_{i=1}^{L}$ on this support. Each model $v_{\boldsymbol{\theta}}$ operates in a latent space $\boldsymbol{z}$ and is trained to remove Gaussian noise $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ added to the target latent $\boldsymbol{z}_0$, *i.e.* $\boldsymbol{z}_t = (1-t)\boldsymbol{z}_0 + t\boldsymbol{\epsilon}$, where $t \sim \mathcal{U}(0,1)$ is a random interpolation timestep. Formally, each model $v_{\boldsymbol{\theta}}$ is trained to predict the rectified-flow velocity:

$$\mathcal{L}(\boldsymbol{\theta}) = \mathbb{E}_{(\boldsymbol{z}_0, \mathbf{c}_p), t, \boldsymbol{\epsilon}} \left[\| v_{\boldsymbol{\theta}}(\boldsymbol{z}_t, \mathbf{c}_p, t) - (\boldsymbol{\epsilon} - \boldsymbol{z}_0) \|_2^2 \right], \tag{3}$$

where $\mathbf{c}_p$ denotes the encoded partial point observations. The loss is applied independently to both stages, and we fine-tune the newly introduced conditioning modules and the denoisers from the pretrained generative prior. The warm start of Equation (1) and the mask concatenation are active during both training and inference. The metric-scale model is trained on windows: each sample draws a level (weighted 4:2:1:1 from fine to coarse) and a window, crops the depth points of 2–8 views that observe it, and supervises that window's structure or shape latents. The parent of Equation (2) is, with probability 0.7, predicted by an earlier metric-scale model and otherwise the ground-truth parent, randomly dilated, cut or dropped, so that training parents err like those seen at inference.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Evaluation datasets. We evaluate on two indoor datasets with ground-truth geometry for both visible and occluded regions: **Replica** (Straub et al., 2019) with 8 photorealistic indoor reconstructions with high-quality meshes and densely sampled trajectories, and **SCRREAM** (Jung et al., 2024) with 12 real-world captures with accurate per-scene ground truth, which we treat as our primary real-world benchmark.

Training data construction. We train ANCHORCOMPLETE on **ScanNet++** (Yeshwanth et al., 2023) and **InternScenes** (Zhong et al., 2025). ScanNet++ supplies real-world RGB-D videos and scanned meshes, but the meshes are incomplete and not watertight. In contrast, we use around 16k scenes from InternScenes whose watertight glTF meshes serve as complete supervision $\mathcal{M}$, covering both observed surfaces and occluded regions. Per scene we render multi-view RGB-D with BlenderProc (Denninger et al., 2023) at 640×480 and an 85° horizontal field of view, placing cameras with an area-adaptive multi-anchor sampler plus corner viewpoints for coverage. Details are in the appendix.

Metrics. Following prior work (Wang et al., 2024; 2025a; Chen et al., 2026b), we report Chamfer Distance (CD) and F-score, using ground-truth poses to crop the **visible** mesh within the input view frustums. We decompose CD into *accuracy* (prediction-to-GT) and *completeness* (GT-to-prediction). $\mathrm{CD}_{\mathrm{vis}}$ is symmetric over the visible region, while $\mathrm{CD}_{\mathrm{all}}$ combines visible-region accuracy with full-scene completeness. F-scores at 0.1 and $0.05\,\mathrm{m}$ use the same regions as $\mathrm{CD}_{\mathrm{all}}$; lower CD and higher F are better. Accuracy is computed only for predictions inside the input-view frustums, since completion outside the observed region is inherently multi-modal and the ground truth represents only one valid realization. See Appendix A.2 for details.

Baselines. We compare against three groups, grouped as in Table 1. *Feed-forward dense reconstruction*: VGGT (Wang et al., 2025a), DA3 (Lin et al., 2025), CUT3R (Wang et al., 2025b), and PI-3 (Wang et al., 2025c) predict dense geometry from images but target visible surfaces; none outputs a mesh, so we convert all of them with the same pipeline. *Image-based generation*: TRELLIS (Xiang et al., 2024) and GenRC (Li et al., 2024) test whether generative priors alone recover missing geometry. *Scene completion*: SG-NN (Dai et al., 2020), NKSR (Huang et al., 2023), and Seen2Scene (Meng et al., 2026) take partial point clouds and predict complete surfaces, matching our input-output setting most closely. We run Seen2Scene with the authors' released code and checkpoints; its input is a TSDF fused (1.1 cm voxels, VDBFusion) from the same DA3 depth maps our model conditions on, and both are scored with the same code after the same Sim(3) alignment.

Implementation details. We initialize ANCHORCOMPLETE from the object-level TRELLIS checkpoints. The point-cloud encoder $\mathcal{E}_{\mathrm{p}}$ adopts weights from NOVA3R (Chen et al., 2026b) and is kept

Table 1: **Quantitative reconstruction evaluation on Replica (Straub et al., 2019) and SCR-REAM (Jung et al., 2024) with $N_{\text{views}} = 4$.** The best result in each metric column is shown in **bold**, and the second best is underlined. * denotes our scene finetune of TRELLIS, with its output aligned to the input cloud; the unstarred row is the released object-pretrained model. Runtime is per-scene reconstruction on an RTX PRO 6000 Max-Q. The blue row is the metric-scale model of Section 3.3 (TRELLIS.2, InternScenes with ScanNet++ mixed in), timed on A40/H100 GPUs and left out of the bold/underline marks; rendered figures use the single cube trained on InternScenes only.

Method	Replica				SCRREAM				Runtime ↓
	CD_{vis} ↓	CD_{all} ↓	$F_{0.1}$ ↑	$F_{0.05}$ ↑	CD_{vis} ↓	CD_{all} ↓	$F_{0.1}$ ↑	$F_{0.05}$ ↑	
Feed-forward dense reconstruction									
VGGT (Wang et al., 2025a)	0.248	0.380	0.381	0.247	0.074	0.225	0.533	0.391	<1s
DA3 (Lin et al., 2025)	**0.048**	0.220	0.611	0.508	**0.052**	0.227	0.607	0.453	<1s
CUT3R (Wang et al., 2025b)	0.244	0.360	0.355	0.214	0.189	0.359	0.455	0.311	<1s
PI-3 (Wang et al., 2025c)	0.053	0.224	0.573	0.426	0.096	0.253	0.485	0.315	<1s
Image-based Generation									
TRELLIS (Xiang et al., 2024)	0.537	0.584	0.239	0.127	0.346	0.416	0.262	0.160	∼4 s
TRELLIS* (Xiang et al., 2024)	0.244	0.288	0.308	0.150	0.221	0.264	0.323	0.170	∼4 s
GenRC (Li et al., 2024)	0.355	0.366	0.161	0.076	0.295	0.332	0.191	0.091	∼6.6 min
Scene completion									
SG-NN (Dai et al., 2020)	0.085	0.214	0.600	0.415	0.111	0.225	0.518	0.308	18 s
NKSR (Huang et al., 2023)	0.057	0.211	0.633	0.527	0.066	0.209	0.614	0.457	1 s
Seen2Scene (Meng et al., 2026)	0.070	0.132	0.777	0.647	0.075	0.167	0.692	0.525	∼10.4 min
ANCHORCOMPLETE (ours)	0.053	**0.071**	**0.814**	**0.682**	0.056	**0.075**	**0.787**	**0.581**	∼20 s
ANCHORCOMPLETE, metric-scale (Section 3.3)	0.055	0.067	0.819	0.684	0.057	0.076	0.786	0.584	∼3 min

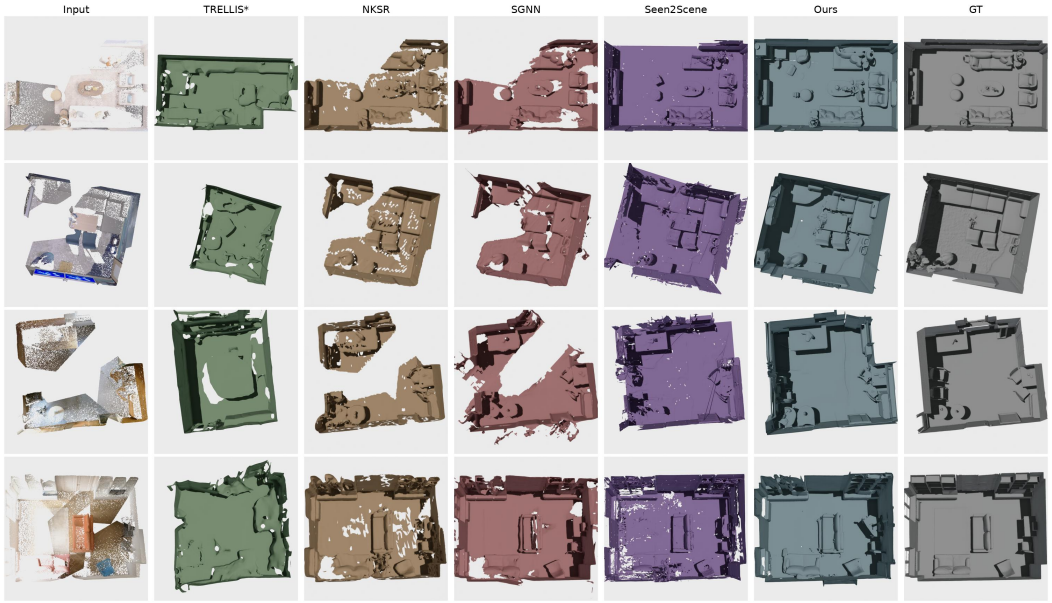


Figure 3: **Qualitative reconstruction results on Replica (Straub et al., 2019) and SCR-REAM (Jung et al., 2024).** Given the same DA3 point cloud as input, our method achieves better scene completion performance.

frozen during training. Only the linear projection head that maps its features to the cross-attention dimension is fine-tuned. The Stage-1 denoiser is a Swin-DiT with 24 blocks, 1024 channels, 16 attention heads, and a window size of 8. It operates on an 8-channel 32^3 sparse-structure latent, which is decoded into a 128^3 occupancy grid. The Stage-2 denoiser uses a sparse Swin-DiT with the same window configuration and operates on the 128^3 structured-latent grid. Both stages are trained independently using AdamW with learning rate 5×10^{-5}. Stage 1 and Stage 2 are each trained for 50k steps on four NVIDIA L40 GPUs. Additional details are provided in the appendix.

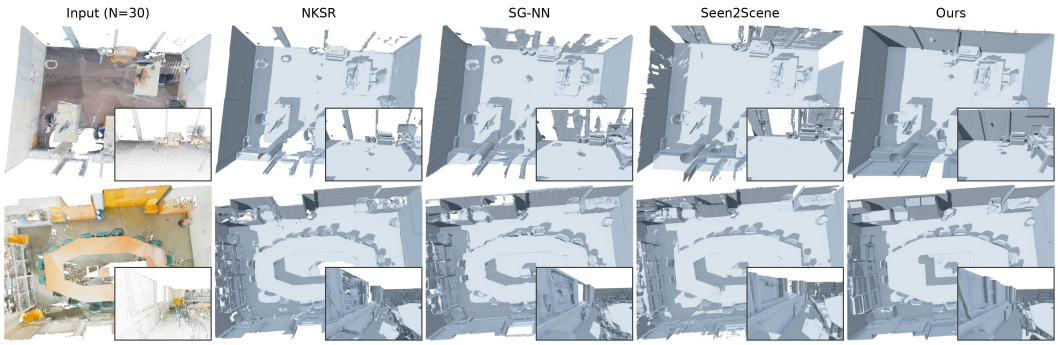


Figure 4: **Real-world scenes.** Two ScanNet++ rooms from thirty views, top-down with an inside view inset (same camera in every column); NKSR (Huang et al., 2023), SG-NN (Dai et al., 2020), Seen2Scene (Meng et al., 2026) (a TSDF fused from the same depth frames) and ANCHORCOMPLETE take the same input. The baselines leave surfaces holed where coverage thins, while ours closes them.

4.2 COMPLETE SCENE RECONSTRUCTION

Table 1 evaluates complete-scene reconstruction at N_{views}=4. ANCHORCOMPLETE leads the complete-region metrics on both datasets against every baseline. On the visible region DA3 (Lin et al., 2025) obtains the best CD_{vis} with ANCHORCOMPLETE close behind, so the completion gains do not cost visible-surface fidelity; VGGT (Wang et al., 2025a) and CUT3R (Wang et al., 2025b) struggle even there, likely because four views give little coverage of a full room. Surface fitting such as NKSR (Huang et al., 2023) smooths visible points into surfaces but still lags, showing smoothing alone cannot recover unseen geometry, while TRELLIS (Xiang et al., 2024) without partial-point conditioning drifts off the captured scene. TRELLIS*, our scene-level fine-tuned version of the same prior, improves substantially over the released object model but still trails ANCHORCOMPLETE by a wide margin on both datasets, indicating that our advantage comes from conditioning on the observed geometry rather than from scene-level training.

The blue row of Table 1 is ANCHORCOMPLETE on TRELLIS.2 latents (Xiang et al., 2025) with the metric-scale levels of Section 3.3, rooted in a single normalized cube ("Ours (TRELLIS.2)" in Table 2(a)); both take the same DA3 input and normalization cube and are trained on InternScenes with ScanNet++ mixed in. Training data, not the backbone, sets accuracy here: on InternScenes alone the single cube reaches CD_{vis} 0.078 and $\text{F}_{0.05}$ 0.620 on Replica, as many extra InternScenes steps leave it at 0.076 and 0.609, and mixing in ScanNet++ brings it to 0.054 and 0.701, on par with the TRELLIS-based model. The metric-scale levels keep these numbers and raise $\text{F}_{0.02}$ on Replica from 0.235 to 0.251. This protocol favours a single cube, whose rooms are small and whose cube comes from the ground-truth bounding box; Table 2(b) and Figure 8 show what the finer levels add on larger held-out ScanNet++ scenes.

Figure 3 shows the same split visually. ANCHORCOMPLETE keeps the visible surface and closes back walls and out-of-view floor where the feed-forward and surface-fitting baselines leave holes. TRELLIS* returns a plausible room unrelated to the input cloud. Seen2Scene (Meng et al., 2026), the completion baseline closest to our setting, returns a far flatter result, without the furniture relief and wall structure ANCHORCOMPLETE preserves. It is also the slowest method in Table 1, at $\sim$10.4 min per scene against our $\sim$20 s.

Real-world scenes. On two ScanNet++ rooms reconstructed from thirty views (Figure 4), all methods recover the layout, but the baselines leave surfaces ragged and holed where coverage thins while ours closes them; with no complete ground truth, these rooms are shown qualitatively.

4.3 ROBUSTNESS TO SPARSE INPUT VIEWS

To evaluate robustness to sparse input views, we uniformly subsample each scene's saved views to $N_{\text{views}} \in \{2, 4, 6, 8, 10\}$ and report the same metrics as in Table 1, comparing against DA3 (Lin et al., 2025) and NKSR (Huang et al., 2023) as representative baselines. ANCHORCOMPLETE leads CD_{all} and the F-scores at every view count we test, against DA3 and NKSR, with the largest gains in the

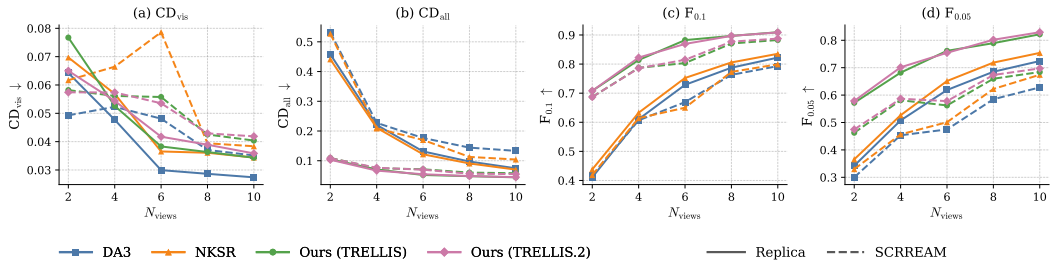


Figure 5: **Sparse-view robustness on Replica (Straub et al., 2019) and SCRREAM (Jung et al., 2024)** (mean across scenes). Ours (TRELLIS): the TRELLIS-based model of Table 1; Ours (TRELLIS.2): the TRELLIS.2 single cube trained with ScanNet++ mixed in.

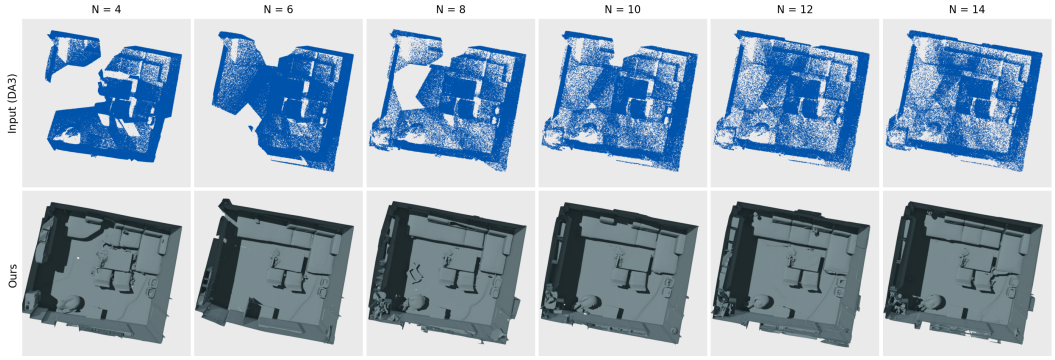


Figure 6: **Sparse-view conditional scene completion.** The DA3 input (top) loses coverage as views are removed, while our completion (bottom) stays closed across the range. Ours is the TRELLIS.2 single cube; N=12 and 14 add held-out frames, as Replica provides 10 input frames per scene.

sparsest setting: the gap to NKSR in CD_{all} is widest at N=2 on both datasets. It also degrades less as views decrease, maintaining higher F-scores from N=10 to N=2 than visible-surface baselines. The TRELLIS.2 single cube (pink in Figure 5) also leads CD_{all} and $F_{0.05}$ at every view count on both datasets. CD_{vis} remains slightly better for DA3, especially at very sparse views, since it reconstructs only observed surfaces while our model generates complete geometry. Figure 6 shows that, as views are removed, the DA3 cloud becomes increasingly incomplete whereas our reconstruction remains complete. In addition, Figure 7a shows that AnchorComplete can generate diverse plausible completions from the same partial observation, suggesting that it does not simply memorize a single dataset-specific shape prior.

4.4 Ablation Study

We ablate on Replica (Straub et al., 2019) to analyze the contributions of key design choices in AnchorComplete.

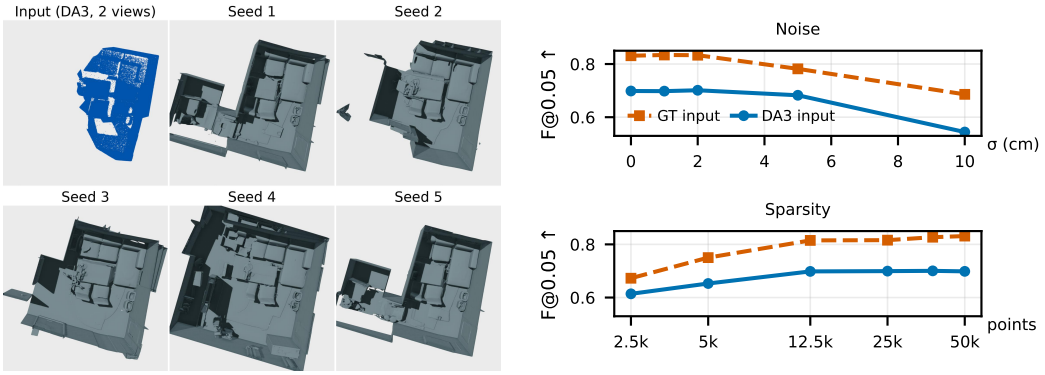
Table 2: **Ablation study.** (a) Partial-point conditioning on Replica (Straub et al., 2019): CD in metres (↓), F-scores (↑) at 0.1 / 0.05 / 0.02 m. N_{views}=4, 8 scenes; Global is global anchor tokens and Local is local spatial injection. (b) The finest octree level of the metric-scale model on held-out scenes. Both conditioning paths are needed, and only the finest level recovers detail at 2 cm, at a higher runtime.

(a) **Partial-point conditioning**

Method	Global	Local	Backbone	CD_{vis}	CD_{all}	$F_{0.1}$	$F_{0.05}$	$F_{0.02}$
TRELLIS*	✗	✗	DiT (16^3)	0.244	0.288	0.308	0.150	0.030
w/o local injection	✓	✗	Swin-DiT (32^3)	0.072	0.105	0.749	0.532	0.131
w/o global tokens	✗	✓	Swin-DiT (32^3)	0.060	0.086	0.787	0.652	0.219
Ours	✓	✓	Swin-DiT (32^3)	**0.053**	0.071	0.814	**0.682**	**0.245**
Ours (TRELLIS.2)	✓	✓	DiT (16^3)	0.054	0.067	0.822	0.701	0.235

(b) **Octree depth**

Finest	Cell	Acc	Comp	$F_{0.05}$	$F_{0.02}$	$F^{det}_{0.02}$	$R_{0.1}$	Time (s)
$\ell = 2$	1.5–3.4 cm	0.045	0.046	0.835	0.485	0.429	0.900	**20**
$\ell = 1$	1.6 cm	0.038	**0.036**	**0.873**	0.675	0.526	**0.919**	99
$\ell = 0$ (ours)	0.78 cm	**0.028**	0.037	0.872	**0.801**	**0.673**	0.900	260



(a) (b)

Figure 7: **Generation diversity and input robustness. (a)** From one partial point cloud, different seeds give different plausible scenes; the observed structure is preserved and the variation sits in unobserved regions. **(b)** Robustness to input noise and sparsity, the noise sweep at a fixed 50k points, each run on the DA3 cloud and on ground-truth points. (a) renders the TRELLIS.2 single cube trained on InternScenes only; (b) plots the single cube trained with ScanNet++ mixed in.

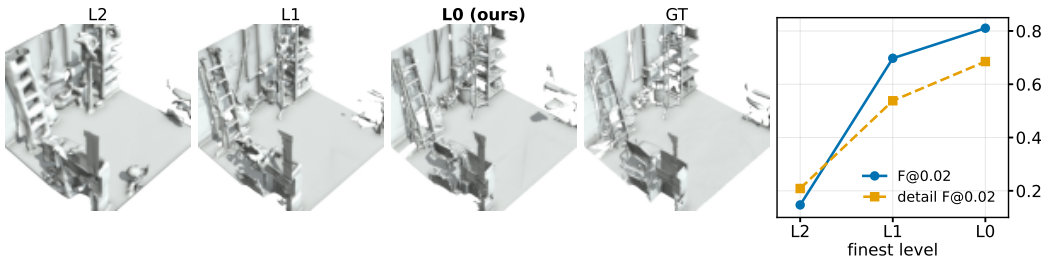


Figure 8: **Octree depth.** The same model stopped at levels 2, 1 and 0 on a room corner of ScanNet++ scene de3c77cecd, with the mean F-scores of Table 2(b) on the right. Only level 0 separates the ladder rungs and the shelves of the metal rack.

Importance of Partial-point Conditioning. Partial-point conditioning anchors the generative prior through two complementary paths (Section 3.2): *global anchor tokens*, encoded from the input cloud and read via cross-attention, and *local spatial injection*, which injects the visible cloud into the structure latent. Removing local injection causes the largest drop, especially in visible-region fidelity ($\mathrm{CD}_{\mathrm{vis}}$), since per-voxel evidence localizes the observed geometry. Removing global anchor tokens causes a smaller drop, as the tokens supply scene-level context rather than localization.

Octree Depth. Table 2(b) and Figure 8 stop the same model at a coarser level on five held-out ScanNet++ scenes and one Matterport3D house (ground-truth depth from 10 views, scored inside the region seen by 30 views; $\mathrm{F}^{\mathrm{det}}_{0.02}$ is the F-score at 2 cm on high-curvature ground truth). Level 2 is the single cube, whose cell grows with the scene. The gain appears once the cell falls below the 2 cm tolerance: $\mathrm{F}_{0.02}$ is 0.485 at level 2, 0.675 at level 1 and 0.801 at level 0, which also adds most of the detail ($\mathrm{F}^{\mathrm{det}}_{0.02}$ 0.526 to 0.673) and lowers the accuracy error from 0.038 to 0.028 m. Recall at 10 cm is set by the coarse levels (0.900 to 0.919); the cost is time, 20, 99 and 260 s per scene on one H100.

Robustness to Imperfect Inputs. All results above condition on raw DA3 clouds with no confidence filtering. We additionally perturb that input, with no retraining (Figure 7b). $\mathrm{F}_{0.05}$ is unchanged down to 12.5k of the 50k points and drops 0.085 at 2.5k, whereas noise is harmless up to $\sigma = 2$ cm, the voxel scale, and costs 0.155 by $\sigma = 10$ cm, as the model relies on spatially accurate anchors. Both sweeps keep their shape on ground-truth points as well as DA3, so the curves show the model's response to the perturbation, not an interaction with DA3's own error.

5 CONCLUSION

We presented ANCHORCOMPLETE, a generative framework for recovering complete scene meshes from partial point clouds. Our approach combines partial-point conditioning with metric-scale coarse-to-fine generation on an octree of fixed-size windows, which keeps the mesh resolution fixed in metres regardless of scene extent and generates complete scene geometry directly in the original scene coordinate frame. ANCHORCOMPLETE improves complete-scene reconstruction over the baselines while maintaining high fidelity to the observed geometry. We further demonstrate robustness to sparse-view inputs and strong performance on real-world scenes, highlighting its potential for applications that require complete digital representations of physical environments.

AI USE STATEMENT

We did not use generative AI tools for research ideation, methodology design, or as an important, original, or non-standard component of the core methods in this work. We used AI-assisted writing tools for grammar and phrasing suggestions during manuscript preparation; all such suggestions were reviewed and edited by the authors, who take responsibility for the final content of this work.

ETHICS STATEMENT

This work addresses complete scene-mesh reconstruction from partial point clouds, evaluated on publicly available research datasets (ScanNet++, InternScenes, Replica, SCRREAM). We are not aware of ethical concerns specific to this research, including risks to human subjects, dataset consent issues, or foreseeable dual-use harms beyond those already associated with general-purpose 3D reconstruction and generative modeling research.

REPRODUCIBILITY STATEMENT

We describe the model architecture, training data construction, and evaluation protocol in Sections 3 and 4 of the main text, with the camera sampler in Appendix A.1 and the remaining implementation details, metric definitions, and evaluated scene subsets in Appendix A.2. We plan to release code and pretrained checkpoints upon publication to support independent verification of the reported results.

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A Appendix

A.1 Training Data Construction Details

For each `InternScenes` room we place 1–9 camera anchors by floor area, clear of furniture, and render three pitch rings at $-10°$, $-45°$, and $+50°$ from each, with 10 uniformly spaced yaw angles per ring, giving 30 frames per anchor. Zenith and nadir views and corner cameras facing the room interior cover the boundaries that centre-facing views miss. The sampler yields 40 to roughly 420 frames per scene depending on room size, 106 on average; training draws random subsets of these to form the partial point cloud.

A.2 More Implementation Details

Stage-1. The Stage-1 denoiser operates on the $32^3 \times 8$ TRELLIS sparse-structure latent, with windowed self-attention across the two levels of a UNet and adaptive layer-norm time conditioning. Cross-attention attends to 768 tokens produced by the frozen NOVA3R encoder from the 50k-point input cloud. Two modifications make it visibility-aware: (i) a *warm-started prior* that voxelizes the input cloud at 128^3, runs the frozen TRELLIS SS-encoder to obtain $z_{\text{vis}} \in \mathbb{R}^{8\times 32^3}$, and replaces the Gaussian initial noise at visible voxels by z_{vis}; and (ii) a VisibleMaskConcat adapter, which concatenates the binary visibility mask with the noisy latent at each denoising step and maps the result back to 8 channels with an identity-initialized 3D convolution. After 50 flow-sampling steps with classifier-free guidance, the latent is decoded by the unmodified TRELLIS sparse-structure decoder into a 128^3 occupancy field, whose active voxel coordinates are exported to Stage 2.

Stage-2. The Stage-2 denoiser is *sparse*, operating directly on the active-coord tensor produced by Stage 1. Tokens are formed by patchifying the 128^3 sparse grid with patch size 2, giving an effective 64^3 token grid sparsified to the active set. Positional encoding is local-RoPE inside each Swin window rather than a global APE, so the same operator applies to scenes of differing spatial extent. Cross-attention reuses the Stage-1 tokens. After 50 flow steps with CFG 3.0, the predicted SLAT is fed to the unmodified TRELLIS mesh decoder.

Metric definitions. Let $\mathcal{P}$ be points sampled from the predicted mesh, $\mathcal{G}$ points sampled from the complete ground-truth mesh, and $\mathcal{G}_{\text{vis}}$ the ground-truth points returned by raycasting the input cameras, so $\mathcal{G}_{\text{vis}}$ is occlusion-aware. Let $\mathcal{P}_{\text{vis}}$ be the subset of $\mathcal{P}$ inside the union of the input view frustums; this is a geometric frustum test, not an occlusion test, since occlusion of the prediction cannot be resolved without raycasting it. With $d(x, \mathcal{S}) = \min_{y\in\mathcal{S}}\|x - y\|$,

$$a = \frac{1}{|\mathcal{P}_{\text{vis}}|}\sum_{x\in\mathcal{P}_{\text{vis}}} d(x,\mathcal{G}), \qquad c_{\text{vis}} = \frac{1}{|\mathcal{G}_{\text{vis}}|}\sum_{y\in\mathcal{G}_{\text{vis}}} d(y,\mathcal{P}), \qquad c_{\text{all}} = \frac{1}{|\mathcal{G}|}\sum_{y\in\mathcal{G}} d(y,\mathcal{P}),$$

$$\text{CD}_{\text{vis}} = \tfrac{1}{2}(a + c_{\text{vis}}), \qquad \text{CD}_{\text{all}} = \tfrac{1}{2}(a + c_{\text{all}}), \qquad \text{F}_\tau = \tfrac{2P_\tau R_\tau}{P_\tau + R_\tau},$$

with $P_\tau = |\{x \in \mathcal{P}_{\text{vis}} : d(x,\mathcal{G}) < \tau\}|/|\mathcal{P}_{\text{vis}}|$ and $R_\tau = |\{y \in \mathcal{G} : d(y,\mathcal{P}) < \tau\}|/|\mathcal{G}|$. Precision is therefore restricted to the observed region while recall covers the scene, so geometry generated outside the frustums is not charged. We sample 10^5 points per mesh.

Evaluation Datasets. For Replica (Straub et al., 2019), we use 8 of the 18 scenes, following Ni et al. (2025). For SCRREAM (Jung et al., 2024), we exclude scenes whose videos focus only on a single corner of the scene.

A.3 Additional Qualitative Results

More reconstructions on Replica and SCRREAM. We provide additional top-down qualitative reconstructions on Replica (Straub et al., 2019) and SCRREAM (Jung et al., 2024) at N_{views}=4 in Figures 9 and 10, respectively, comparing the input partial point cloud against representative baselines and AnchorComplete. Across both datasets, our method recovers a complete room mesh, including floors, walls, and large objects in regions where the partial input has no coverage, while baselines leave visible holes wherever the cameras did not directly observe.

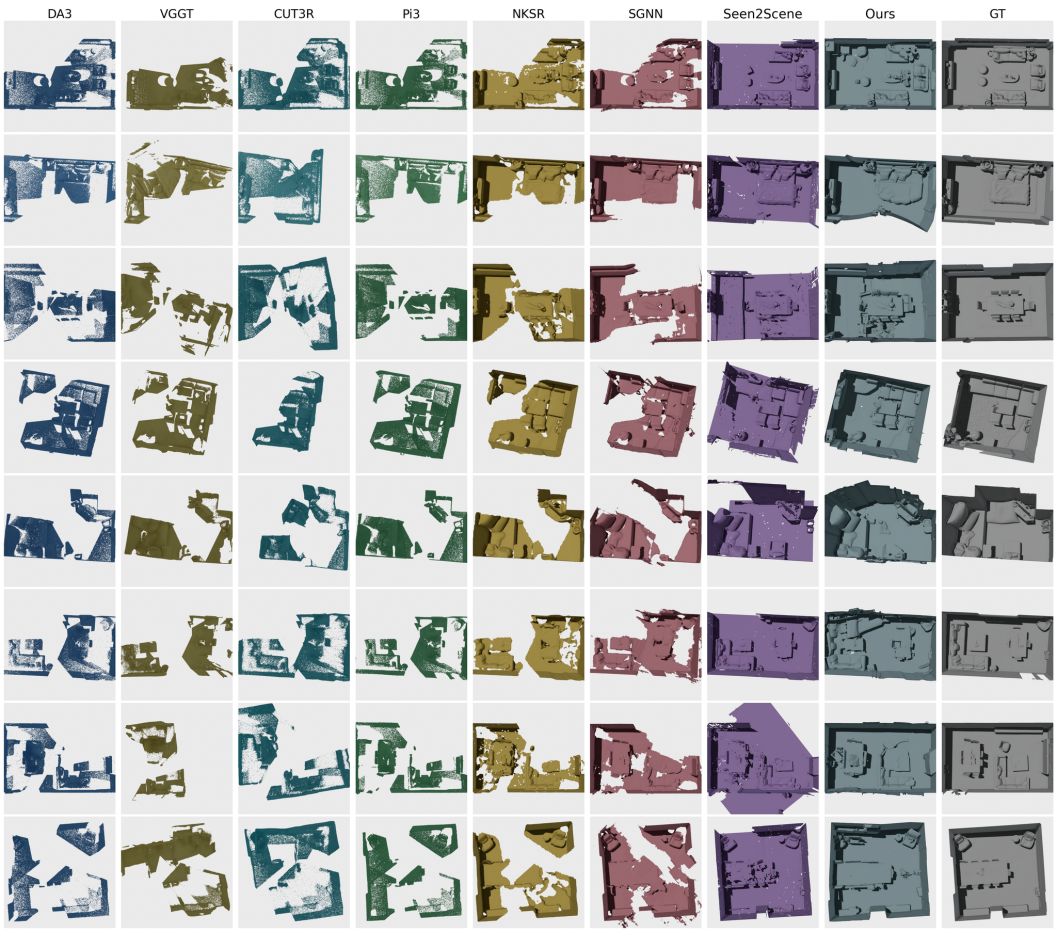

Figure 9: **Additional qualitative results on Replica (Straub et al., 2019) at $N{=}4$.** Top-down views of the meshes each method reconstructs from the same partial DA3 point cloud, with the ground truth in the last column.

A.4 Generalization Across Input Sources

The experiments above condition on point clouds produced by DA3 (Lin et al., 2025), a per-view point-map estimator. Because AnchorComplete conditions directly on points rather than on images or estimator-specific features, it is not architecturally tied to this source. To test this, we apply the same pretrained checkpoint, with no retraining or source-specific tuning, to partial point clouds from four acquisition paradigms: learned per-view point maps (DA3, Pi3 (Wang et al., 2025c), VGGT-Ω (Wang et al., 2026)), classical structure-from-motion and multi-view stereo (COLMAP (Schönberger & Frahm, 2016)), monocular visual SLAM (DROID-SLAM (Teed & Deng, 2021), DPVO (Teed et al., 2022)), and a simulated 32-beam LiDAR scan. All inputs are registered into the same evaluation frame and scored with the protocol used for the default DA3 setting.

Table 3 shows that AnchorComplete remains effective across every source we test, with $F_{0.05}$ ranging from 0.555 to 0.761. VGGT-Ω is the weakest of them and DROID-SLAM the strongest; the same checkpoint handles both without tuning. In particular, the SfM/MVS, visual-SLAM, and LiDAR rows confirm that the method is not restricted to the per-view point-map setting used elsewhere in the paper.

A.5 Symmetric Full-Scene Chamfer

The accuracy term of Table 1 counts only predicted points inside the input frustums, so geometry generated in unobserved space is never charged. Table 4 repeats CD_{all} with accuracy taken over all

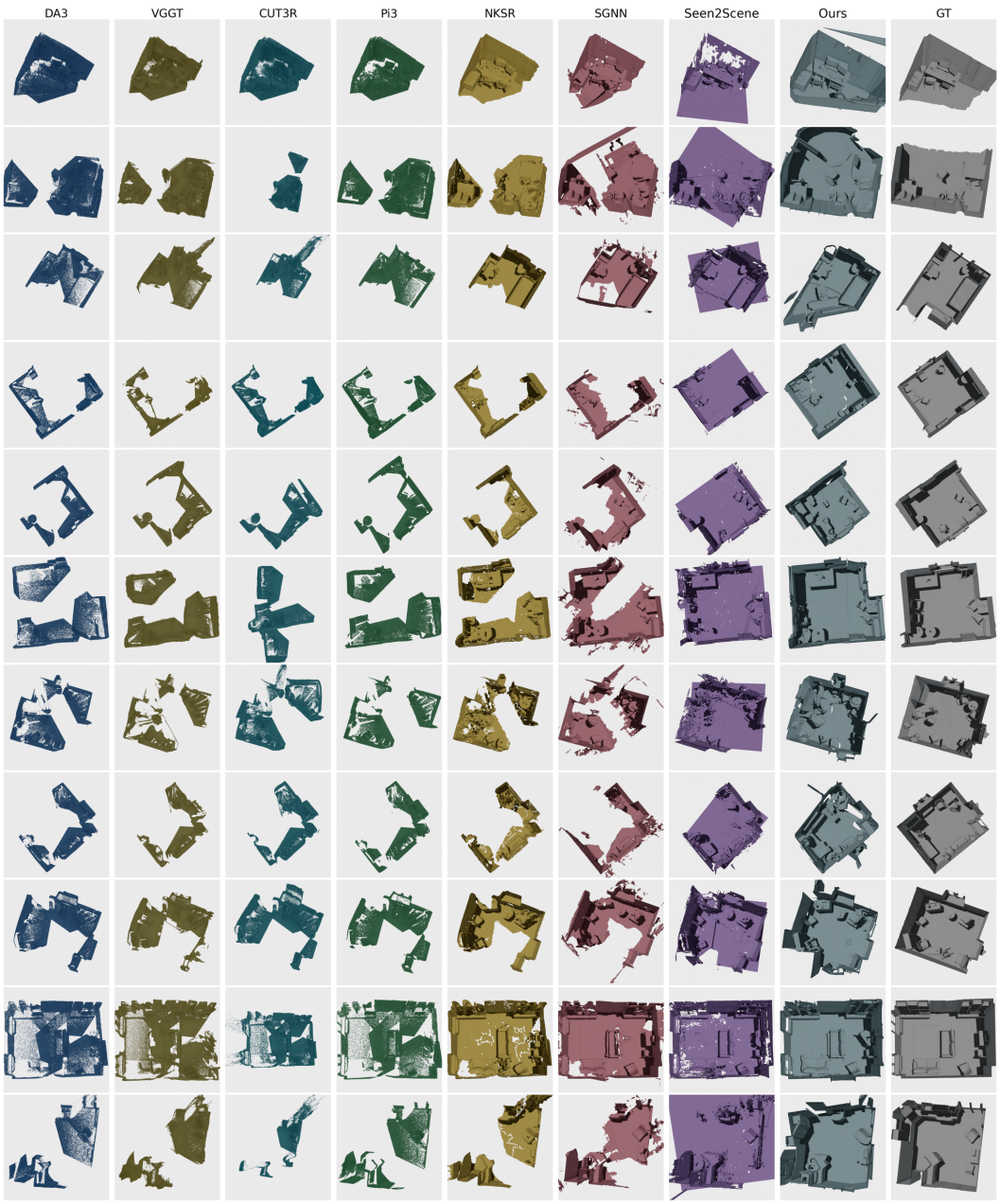

Figure 10: **Additional qualitative results on SCRREAM (Jung et al., 2024) at N=4.** Top-down views of the meshes each method reconstructs from the same partial DA3 point cloud, with the ground truth in the last column.

predicted points instead. The baselines barely move, since they place almost no geometry outside the frustums; ours rises, which is what generating into unobserved space costs. The ordering is unchanged on both datasets. We keep the asymmetric form as the headline because a completion that differs from the recorded ground truth in unobserved space is not thereby wrong.

A.6 Failure Cases

Figure 7b perturbs the point cloud synthetically, which is not how a depth estimator actually fails. To probe that, we degrade the *images* DA3 consumes and re-run the full pipeline on the resulting clouds. Figure 11 shows a SCRREAM scene under low light and a ScanNet++ scene under motion blur and

Table 3: **Generalization across input point-cloud sources on Replica (Straub et al., 2019).** The same pretrained AnchorComplete checkpoint is evaluated without retraining on partial point clouds from multiple acquisition paradigms, using the default DA3 evaluation protocol (Table 1). Visual-SLAM and LiDAR use full trajectories/scans and therefore demonstrate source generality rather than budget-matched performance. † LiDAR uses conservative 5 cm range noise. The right block runs the TRELLIS.2 single-cube AnchorComplete on exactly the same input clouds.

		Ours (TRELLIS)				Ours (TRELLIS.2 single cube)			
Category	**Input source**	$CD_{vis}\downarrow$	$CD_{all}\downarrow$	$F_{0.1}\uparrow$	$F_{0.05}\uparrow$	$CD_{vis}\downarrow$	$CD_{all}\downarrow$	$F_{0.1}\uparrow$	$F_{0.05}\uparrow$
Per-view point map	DA3 (default)	0.053	0.071	0.814	0.682	0.054	0.067	0.822	0.701
	Pi3	0.047	0.067	0.822	0.641	0.048	0.062	0.837	0.655
	VGGT-Ω	0.066	0.082	0.748	0.555	0.068	0.079	0.746	0.547
SfM/MVS	COLMAP	0.067	0.085	0.760	0.618	0.069	0.084	0.770	0.641
Visual SLAM	DROID-SLAM	0.043	0.055	0.872	0.761	0.044	0.053	0.870	0.759
	DPVO	0.048	0.060	0.863	0.682	0.046	0.055	0.872	0.713
LiDAR	32-beam, σ=5 cm†	0.040	0.044	0.942	0.744	0.037	0.043	0.949	0.761

Table 4: **Charging for generated geometry** (N_{views}=4, metres ↓). CD_{all} is the value reported in Table 1, whose accuracy term counts only predicted points inside the input frustums; CD_{all}^{sym} takes accuracy over all predicted points.

Method	**Replica**		**SCRREAM**	
	CD_{all}	CD_{all}^{sym}	CD_{all}	CD_{all}^{sym}
DA3 (Lin et al., 2025)	0.220	0.220	0.227	0.228
NKSR (Huang et al., 2023)	0.211	0.211	0.209	0.208
Seen2Scene (Meng et al., 2026)	0.132	0.134	0.167	0.175
AnchorComplete (ours)	**0.071**	**0.098**	**0.075**	**0.110**
AnchorComplete, TRELLIS.2 single cube	0.067	0.084	0.076	0.114

sensor noise, both simulated on real captures. The SCRREAM scene is held out; the ScanNet++ one is in our Stage-1 training set, so the low-light row carries the weight of the claim.

The degraded imagery costs DA3 large parts of the room, and the reconstruction inherits exactly those gaps: AnchorComplete still closes the surface it is given and produces a coherent room, but it completes around the missing observations rather than recovering them. The model is trained only on clean geometry, so it treats the input as a reliable anchor and preserves systematic upstream error rather than repairing it. Training with degraded inputs is a natural remedy we leave for future work.

A.7 Limitations

While AnchorComplete is robust to sparse input views (Section 4.3), it still has several limitations. First, the current model predicts geometry only: we use the inherited mesh decoder and do not model appearance. Extending the framework with RGB supervision and an appearance-aware decoder could enable textured mesh generation. Second, performance depends on the quality of the upstream geometry, so drift, scale errors, or systematic depth bias may propagate to the generated mesh. Appendix A.6 shows the sharper form of this: severe upstream failures are inherited rather than corrected. Third, our training and evaluation focus on indoor scenes, and generalization to outdoor, unbounded, or substantially different scene distributions remains open.

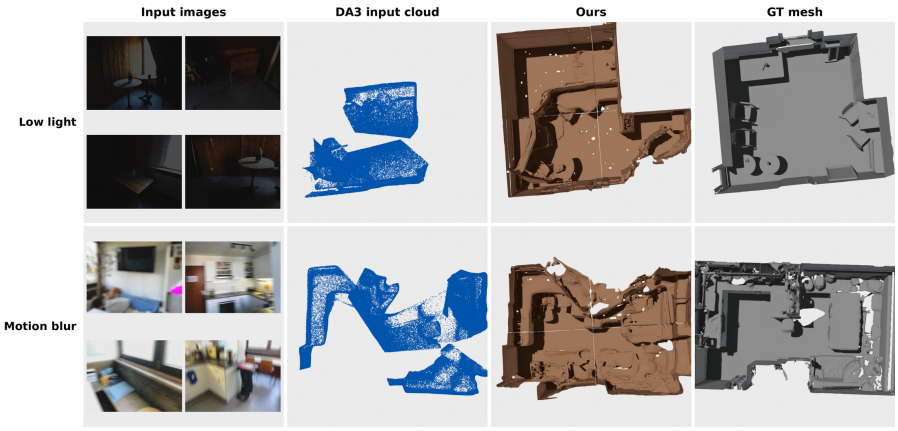

Figure 11: **Failure cases under degraded imagery.** Real captures with simulated low light (top) and motion blur/noise (bottom). The degraded inputs cause DA3 to miss large scene regions, and our model completes around these gaps rather than recovering them.